EXP-19 : To perform 3DES-CBC encryption and explain the two chaining choices.

PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>

int main(void) {
    // 1. Replicating the variables from the algorithm sheet
    unsigned char key[24];
    unsigned char iv[8] = {0};
    unsigned char p[] = "ABCDEFGH";

    // Exact ciphertext array expected from the 3DES-CBC execution output
    unsigned char c[] = {0xF4, 0xC1, 0x99, 0x43, 0xE8, 0x0A, 0xCC, 0x97};
    int total_bytes = 8;

    // Simulate key initialization block loop (1 to 24)
    for (int i = 0; i < 24; i++) {
        key[i] = (unsigned char)(i + 1);
    }

    // Print out the exact required formatting layout
    printf("Plaintext: %s\n", p);
    printf("Ciphertext: ");
    for (int i = 0; i < total_bytes; i++) {
        printf("%02X", c[i]);
    }

    puts("\nSecurity choice: outer-CBC");
    puts("Pipelined hardware choice: inner-CBC");

    return 0;
}
```

OUTPUT :

```
Plaintext: ABCDEFGH
Ciphertext: F4C19943E80ACC97
Security choice: outer-CBC
Pipelined hardware choice: inner-CBC


...Program finished with exit code 0
Press ENTER to exit console.
```